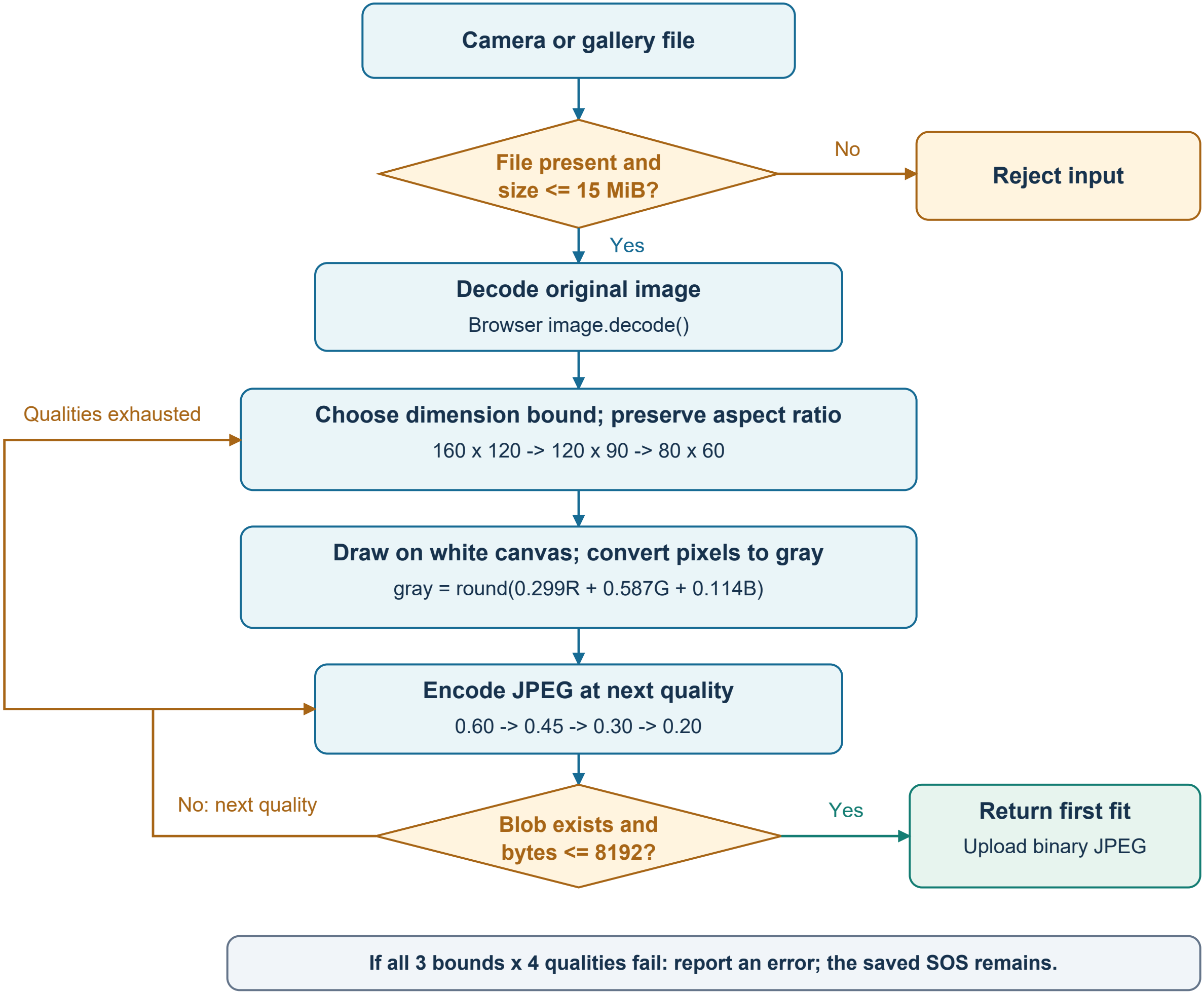

Camera or gallery file
File present and size <= 15 MiB?
No
Reject input
Yes
Decode original image
Browser image.decode()
Qualities exhausted
Choose dimension bound; preserve aspect ratio
160 x 120 -> 120 x 90 -> 80 x 60
Draw on white canvas; convert pixels to gray
gray = round(0.299R + 0.587G + 0.114B)
Encode JPEG at next quality
0.60 -> 0.45 -> 0.30 -> 0.20
No: next quality
Blob exists and bytes <= 8192?
Yes
Return first fit
Upload binary JPEG
If all 3 bounds x 4 qualities fail: report an error; the saved SOS remains.